

DS701 Homework 3: k-means and Linkage

Released Mon Sep 21 · due Mon Sep 28, 11:59 pm on Gradescope

Name: _____ BU ID: _____

How this works. Answer **by hand on paper** (or a tablet with a stylus), show your work, then scan and upload to Gradescope, matching each problem to its pages. This is deliberate practice for the closed-notes quizzes, which are in the same style — so do it without a computer or an AI assistant. It is **participation-graded**: a serious attempt at every problem earns full credit, and worked solutions are released the day after the deadline. Budget 1–2 hours.

Covers: Lecture 03 (Clustering I: k -means) and Lecture 04 (Clustering II: hierarchical).

Problem 1: Lloyd’s algorithm by hand

Practices: running k -means step by step, tracking the objective, and seeing a bad local minimum.

Six one-dimensional data points:

$$x = 0, 5, 6, 7, 12, 15.$$

Cluster them with k -means, $K = 2$, using Euclidean distance (on a line, $\|x - c\|^2 = (x - c)^2$). The objective is the WCSS / inertia, $\sum_k \sum_{x \in C_k} (x - c_k)^2$.

- (a) Start from the centers $c_1 = 12$, $c_2 = 15$. Run **iteration 1**: assign every point to its nearest center, then update each center to the mean of its cluster. Report the two clusters, the WCSS *right after the assignment step* (using the old centers), the new centers, and the WCSS *right after the update step* (using the new centers, same assignment).

Iteration 1	Cluster 1	Cluster 2
Points assigned	_____	_____
WCSS after assignment (old centers)	_____	_____
New centers	_____	_____
WCSS after update (new centers)	_____	_____

- (b) Run **iteration 2** the same way and report the same four things. Then state whether a third assignment step would change anything, and list the WCSS values you have seen, in order. Did the objective ever go up?

Iteration 2	Cluster 1	Cluster 2
Points assigned	_____	_____
WCSS after assignment (old centers)	_____	_____
New centers	_____	_____
WCSS after update (new centers)	_____	_____

Would a third assignment step change anything? _____

WCSS values seen, in order: _____ Ever went up? _____

- (c) Now start instead from $c_1 = 0$, $c_2 = 5$. Run Lloyd's algorithm until the centers stop changing (it is short). Report the final clusters, centers, and WCSS. Compare with (b): which run found the better clustering, and is the other run "wrong" from the algorithm's point of view?

Final clusters: _____ Centers: _____ WCSS: _____

- (d) Two sentences: what does (c) imply for how k -means is run in practice, and what would the k -means++ idea from the lecture have done with this data? (Check your claim: run one iteration from the two centers k -means++ would most plausibly pick.)

Problem 2: Why k-means stops, and why more clusters always “help”

Practices: the two one-line facts behind Lloyd’s algorithm, and reading the WCSS correctly when choosing K .

Write $J(C, c) = \sum_{k=1}^K \sum_{x \in C_k} \|x - c_k\|^2$ for the objective, where $C = (C_1, \dots, C_K)$ is the assignment of points to clusters and $c = (c_1, \dots, c_K)$ the centers.

- (a) **Assignment step.** Hold the centers c fixed and let C' be the assignment that sends every point to its nearest center. Show that $J(C', c) \leq J(C, c)$ for *any* assignment C . (One or two lines: compare the two sums term by term.)
- (b) **Update step.** Hold an assignment fixed and look at one cluster with points $x_1, \dots, x_n$ (one-dimensional, to keep the algebra short — the same argument works coordinate by coordinate). Show that $f(c) = \sum_{i=1}^n (x_i - c)^2$ is minimized at $c = \bar{x} = \frac{1}{n} \sum_i x_i$. Either differentiate f and set the derivative to zero (and say why that critical point is a minimum), or expand $(x_i - c)^2 = ((x_i - \bar{x}) + (\bar{x} - c))^2$ and simplify.
- (c) **Termination.** Using (a), (b), and one further fact — that there are only finitely many ways to assign N points to K clusters — explain in a short paragraph why Lloyd’s algorithm must stop after finitely many iterations (assume, as `sklearn` does, that a point stays where it is when it is equally close to two centers). Then say in one sentence what the argument does *not* promise about the clustering it stops at.

- (d) **Choosing K .** Let J_K^* be the smallest WCSS achievable with K clusters (the global optimum, not a particular run). Explain why $J_{K+1}^* \leq J_K^*$ for every K , and what J_N^* is when N is the number of points. Conclude why “pick the K with the smallest WCSS” is not a usable rule, and name the criterion the lecture uses instead — describe in one sentence what it looks for.

$J_N^* =$ _____ Criterion the lecture uses: _____

Problem 3: Single vs. complete linkage on five points

Practices: agglomerative clustering step by step, dendrogram heights, chaining, and matching a linkage to a data shape.

Five objects A–E with pairwise distances (a metric):

	A	B	C	D	E
A	0	2	5	9	12
B	2	0	3	7	10
C	5	3	0	4	8
D	9	7	4	0	5
E	12	10	8	5	0

- (a) Perform agglomerative clustering with **single linkage**. At each of the four steps write the current cluster-to-cluster distances that matter, the pair merged, and the merge height. Then write the result as the 4×4 linkage matrix, rows `[idx1, idx2, dist, count]`, with A–E as indices 0–4 and new clusters numbered 5, 6, ... in creation order.

Single linkage	Pair merged	Height	Linkage-matrix row
Step 1	_____	_____	_____
Step 2	_____	_____	_____
Step 3	_____	_____	_____
Step 4	_____	_____	_____

(b) Repeat with **complete linkage**.

Complete linkage	Pair merged	Height	Linkage-matrix row
Step 1	_____	_____	_____
Step 2	_____	_____	_____
Step 3	_____	_____	_____
Step 4	_____	_____	_____

(c) Sketch both dendrograms (leaves on the x-axis, merge heights on the y-axis). For each, say which two clusters you get by cutting the tree into two flat clusters. Then explain, in terms of the *definitions* of the two linkages, why single linkage swallowed A, B, C, D one at a time while complete linkage kept two compact groups — the lecture’s word for the single-linkage behavior is *chaining*.

Two-cluster cut, single: _____ complete: _____

- (d) For each situation below, name the linkage the lecture would recommend and give one sentence of justification: (i) two crescent-shaped (“half-moon”) clusters that curve around each other; (ii) several roughly round groups of similar size, plus a handful of scattered noise points between them.

Situation (i) — linkage: _____

Situation (ii) — linkage: _____